



SECJ3623: MOBILE APPLICATION PROGRAMMING

Project Name: Study Companion

Project Proposal

Faculty Computing

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1.0 Project Background and Problem Statement

1.1 Project Background

The rapid advancement of mobile technology has significantly influenced the education sector, transforming how communication and academic management take place in schools. Traditional methods of delivering information — such as verbal instructions, paper notices, or fragmented messaging platforms — are no longer sufficient in supporting the dynamic needs of modern students, teachers, and parents. Many schools still rely on manual tracking of homework, inconsistent announcement channels, and limited digital monitoring systems, which often lead to miscommunication and inefficiencies. As education environments become more connected, there is a growing demand for centralized digital solutions that streamline communication and academic task management.

StudyCompanion+ is developed as a mobile-based school management and communication platform designed to bridge this gap. The system integrates task tracking, announcements, performance monitoring, and progress visibility into a single, user-friendly application. By connecting students, teachers, parents, and school administrators in one ecosystem, the application supports better coordination, improves accountability, and enhances transparency in academic activities. This digital approach aligns with the increasing adoption of smart learning environments and aims to promote more organized, responsive, and data-driven school operations.

1.2 Problem Statement

In many schools, academic communication and task management are still handled through verbal reminders, handwritten notes, or multiple uncoordinated platforms. As a result, students frequently forget homework, project deadlines, or important school activities, which negatively affects their academic performance and time management skills. Teachers often need to repeat the same instructions multiple times, increasing their workload and reducing teaching efficiency. Additionally, parents have limited real-time access to their child's academic responsibilities and performance, making it difficult for them to provide timely guidance and support at home.

From an administrative perspective, schools face challenges in obtaining a comprehensive overview of classroom activities, assignment completion rates, and teacher performance. Information is often scattered, delayed, or incomplete, which hinders effective monitoring and decision-making. Without a centralized system, coordination between stakeholders becomes inefficient, leading to communication gaps and reduced accountability. Therefore, there is a clear need for an integrated platform like StudyCompanion+ that unifies communication, task tracking, and performance monitoring into a single system to improve overall school management and learning outcomes.

2.0 Project Objectives and Scope

2.1 Project Objectives

The main objectives of the StudyCompanion are:

- To create a simple and centralized app for managing classroom tasks and reminders.
- To improve communication between teachers, students, and parents.
- To help parents stay informed about their child's tasks and progress.
- To provide school administrators with an overview of classroom activities and teacher performance.
- To encourage better classroom engagement through digital tools.

2.2 Scope

The scope of the StudyCompanion includes:

- Role-based authentication for Students, Teachers, Parents, and School Administrators.
- Task management module for creating, updating, tracking, and submitting assignments.
- Reminder and notification system for deadlines, announcements, and academic activities.
- Student progress dashboard displaying task completion and performance overview.
- Parent access portal to monitor their child's academic progress and receive updates.
- Teacher announcement module for sharing class updates, materials, and important notices.
- Administrator dashboard for managing user accounts, monitoring system activities, and generating reports.
- Secure cloud data storage and real-time synchronization using Firebase Firestore.
- Basic communication feature allowing message exchange between teachers and parents.
- Report generation and data export features for administrative evaluation.

Features such as online payment processing, live video conferencing, and AI-based automated grading are considered out of scope for this project.

3.0 Targeted Users and Identified User Needs

StudyCompanion+ is designed to support multiple stakeholders within the school environment. Each group of users interacts with the system differently and faces unique challenges in managing academic communication and tasks. Identifying these user groups and understanding their specific needs is essential to ensure that the system provides relevant functionalities and solves real problems experienced in schools. The following table summarizes the primary target users and their identified needs.

User Group	Identified User Needs
Students	• A centralized platform to view assignments and deadlines. • Automatic reminders for upcoming submissions and school activities. • A progress overview to track completed and pending tasks.
Teachers	• An efficient system to assign, update, and manage student tasks. • A communication channel to post announcements and share learning materials. • A way to monitor student submissions and overall class performance.
Parents/Guardians	• Visibility into their child's academic progress and responsibilities. • Timely updates and notifications from teachers.
School Administrators	• A centralized system to manage user accounts and monitor system usage. • Reporting tools to analyse task progress and teacher performance.

4.0 Overview of Proposed Application Features

The proposed StudyCompanion+ application is designed to provide an integrated digital platform that enhances communication, task coordination, and academic monitoring within schools. Based on the identified user needs, the system introduces several core features that support students, teachers, parents, and administrators in managing educational activities more effectively. These features aim to reduce miscommunication, improve task completion rates, and provide better visibility into academic performance and school operations.

The following table summarizes the main features of the proposed application and their purposes:

Feature	Description
Role-Based Login System	Provides secure authentication and access control for students, teachers, parents, and administrators based on their roles.
Task Management Module	Allows teachers to create and update assignments, while students can track and submit tasks before deadlines.
Reminder & Notification System	Sends automated alerts for deadlines, announcements, and important academic activities.
Progress Overview Dashboard	Displays task completion status and performance summaries in a visual format.
Parent Access Module	Enables parents to monitor their child's academic progress and receive school updates.
Teacher Announcement Module	Allows teachers to post class updates, reminders, and learning materials
Administrator Dashboard	Provides tools for managing user accounts, monitoring system usage, and generating reports.
Cloud Data Storage	Stores and synchronizes system data securely using cloud technology to ensure reliability and accessibility.

5.0 Project Timeline and Development Milestone

The development of StudyCompanion+ follows an Agile sprint-based approach, where each sprint focuses on delivering specific functional components of the system. This method allows incremental development, continuous testing, and progressive feature integration. The timeline below outlines the key milestones and deliverables for each sprint.

Sprint 1 (13 October 2025 – 26 October 2025)

Focus: Authentication & User Profile

- User authentication and role-based registration (Student, Teacher, Parent, Admin)
- Firebase Authentication setup (email/password)
- Login, logout, and forgot password functionalities
- User profile management (view and edit basic details)
- GitHub repository setup and branching strategy
- Setup of testing framework
- Adoption of BLoC pattern for authentication and profile state management

Sprint 2 (27 October 2025 – 9 November 2025)

Focus: Task Management Module (Core Process #1)

- Task Management CRUD module
- Teachers: Create, update, delete tasks
- Students: View, track, and submit tasks
- Firestore database structure for task collections
- User interface for task list, task details, and submission tracking
- Basic notifications for newly assigned tasks
- BLoC state management for task operations

Sprint 3 (10 November 2025 – 23 November 2025)

Focus: Communication & Announcements (Core Process #2)

- Teacher Announcement Module (CRUD)
- Teachers: Post, edit, delete announcements
- Students & Parents: View announcements
- Parent access to view child's tasks and announcements
- Progress Overview Dashboard
- Students: Completed vs pending tasks
- Teachers: Class progress summary
- Optional lightweight messaging or comment feature
- Push notifications using Firebase Cloud Messaging

Sprint 4 (24 November 2025 – 7 December 2025)

Focus: Reporting, Add-on Features & Deployment

- Administrator dashboard
- User management
- Activity monitoring
- Report generation
- Reports on task completion, student performance, and teacher activity
- Cloud storage integration for attachments (Firebase Storage)
- UI/UX refinements and responsive design improvements
- Final testing (unit testing, integration testing, user acceptance testing)
- Deployment build (APK) and preparation of documentation & presentation

This sprint plan ensures systematic development, where core learning features are completed before administrative and reporting functionalities.

6.0 Conclusion

StudyCompanion+ is proposed as a comprehensive mobile application aimed at improving communication, task coordination, and academic monitoring within school environments. The system addresses common challenges such as missed assignments, repeated instructions, limited parental visibility, and difficulties in administrative monitoring. By integrating students, teachers, parents, and administrators into a single digital platform, the application promotes better collaboration, transparency, and accountability in managing academic activities.

The project adopts an Agile sprint-based development approach to ensure systematic and incremental delivery of system functionalities, beginning with authentication and task management, followed by communication modules and administrative reporting tools. The proposed features, including role-based access, task tracking, notifications, dashboards, and cloud-based data storage, collectively support a more organized and efficient school management process. Overall, StudyCompanion+ has the potential to enhance learning support, reduce communication gaps, and contribute to a more structured and technology-driven educational environment.